



FIRST SEMESTER, 2026–2027
Course Handout Part II

Date: 1st August 2026

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course No : ME G512
Course Title : Finite Element Method
Instructor-in-Charge : Gaurav Watts

1. Course Description:

Fundamental concepts, matrix algebra and gaussian elimination, one-dimensional problems, trusses, two-dimensional problems using constant strain triangles, axisymmetric solids subjected to axisymmetric loading, two-dimensional isoparametric elements and numerical integration, beams and frames, three-dimensional problems in stress analysis, scalar field problems, dynamic considerations, pre-processing and post processing.

2. Scope & Objective of the Course:

The scope of the course is limited to an introduction to FEA as applied to elasticity problems in solid and structural mechanics. This course will train the students to analyze real-world structural mechanics problems using the finite element method. Students will be introduced to the mathematical basis of finite element analysis, on which nearly all structural analysis software are built. The objectives of the course are:

- CO1.** Understand strong and weak formulations, methods of weighted residuals Rayleigh-Ritz method.
- CO2.** Understand the finite element method procedure, which converts a given boundary value problem into a set of algebraic linear equations that can be solved using computer.
- CO3.** Understand the implementation of FEM for linear analysis of 1D and 2D problems of solid mechanics and heat transfer.

3. Textbooks:

- TB1.** *Introduction to the Finite Element Method*, J. N. Reddy, 4th Edition, Tata McGraw-Hill Publishing Company Ltd., 2020

References:

- R1.** *Introduction to Finite Elements in Engineering*, T. R. Chandrupatla and A. D. Belegundu, 5th Edition, Cambridge University Press, 2021

R2. *First Course in the Finite Element Method*, Daryl L Logan, Enhanced Edition, SI Version, Cengage Learning, 2022

R3. *Finite Element Procedures*, K. J. Bathe, Eastern Economy Edition, Prentice Hall India, 2006

R4. *The Finite Element Method for Solid and Structural Mechanics*, O. C. Zienkiewicz, R.L. Taylor, Seventh edition, Oxford : Butterworth-Heinemann, 2014

4. Course Plan:

Weekly plan for 14 weeks with **three lectures per week**

Mod. No.	Module Title	Learning Outcomes	Reference	Lecture No.
1	Introduction to FEM	• Introduction to numerical methods • Significance of FEM	R1-Chapter-1	L1-L2
2	Variational Methods	• Weighted Residual Methods • Boundary Conditions • Weak formulation • Galerkin method, Rayleigh Ritz method • Principle of minimum potential energy	T1-Chapter-2	L3-L6
3	Static Analysis of a Bar using 1D bar element	• Second-order differential equations in one dimension • Derivation of element equations, connectivity of elements, element stiffness matrix for bar element • Assembly of element stiffness matrices • Imposition of boundary conditions • Solution of algebraic equations • Numerical Examples	T1-Chapter-3	L7-L13
4	FE modelling of a Truss	• Derivation of stiffness matrix for 2-noded Planar Truss Element • Imposition of boundary conditions, solution of equations • Numerical Examples	T1-Chapter-6	L14-L17

5	FE modelling of Beams and Frames	• Euler-Bernoulli's Beam theory and Timoshenko's Beam theory • Euler-Bernoulli beam element • Discretization, derivation of element equations, assembly, finite element solution • Stress Analysis of a beam • Shear Locking • FE modeling of a Frame Structure • Numerical examples	T1-Chapter-5	L18-L23
6	FE modelling of Plane Stress and Plane Strain Problems using 2D Triangular Elements	• Plane Stress vs Plane Strain • Constitutive Matrix for Plane stress and Plane Strain • Area coordinates • Interpolation functions for 3-noded and higher triangular elements • Element Stiffness Matrix for CST elements • Numerical Examples	R1-Chapter-6	L24-L29
7	FE modelling of Plane Stress and Plane Strain Problems using 2D Quadrilateral Elements	• Interpolation functions for Isoparametric 4-Noded Quadrilateral Element • Element Stiffness Matrix for 4-noded quad element • Numerical Integration • Numerical Examples	R1-Chapter-8	L30-L34
8	Modal Analysis of Structures	• Hamilton's Principle • Lagrange's Equation of Motion • Mass Matrices for 1D and 2D elements • Numerical Examples	R1-Chapter-11	L35-L37
9	Heat Transfer Problems	• 1D elements, 2D elements • Numerical Examples	R1-Chapter-10	L38-L39
10	Axisymmetric Problems	• Axisymmetric 1D element • Axisymmetric 2D elements • Numerical Examples	R1-Chapter-7	L40-L41

11	Prelude to Advanced Topics	• Drawbacks of FEM • Types of Nonlinearities • Other numerical techniques	Class Notes	L42
----	----------------------------	---	-------------	-----

5. Evaluation Scheme:

Component	Wt (%)	Duration	Date & Time	Comments
Class Quizzes	10	in class	Continuous	Best 10 out of 12
Lab Assignments	20	100 min	Lab Class	Open book, Best 4 out of 5
Project	20	Throughout the Semester	Continuous	Presentations
Mid-Semester Test	20	90 mins.	TBA	Open Book
Comprehensive Examination	30	180 mins.	TBA	Open-Book

NC Criterion: If a student misses majority of the evaluative components or if the student scores less than 15 % of the maximum marks, he/she will be awarded an 'NC' grade.

6. Office Consultation Hour:

Every Wednesday 12pm to 1pm in 2152-Q. Otherwise, please feel free to email me to seek a prior appointment.

7. Make-up Policy:

Make-up tests will be given only to genuine cases. No make-up will be given for the class quizzes, lab assignments and project presentations.

Instructor-in-charge
ME G512 Finite Element Method